



SPARE PART CATALOG

12V4000M60R/526102237



A Rolls-Royce
solution

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This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments

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Preface

This catalog is the basic reference document for ordering spare parts. Illustrations are provided for extensive assembly operations to assist trained personal in conducting such operations but do not claim to be complete. For better overview, even product components that are not available as spare parts are shown.

The individual components were taken from the assembly list.

Amendments service

Part No. changes and amendments which occur after publication of the catalog are not published.

We therefore recommend that users make a record of amendments in their catalogs.

This information is provided, for instance, in order confirmations and on spare parts labels.

Catalog structure

The catalog is divided into Function Groups (FG) and Function Group Segments (FGS) in ascending order according to the MTN 5153 works standard.

Catalog pages

Depending on the type of the product, the header of the page contains the following information:

- **EQUIPMENT TYPE** or **EQUI TYPE** = e.g.: Engine, Gas Turbine, Automation, System; ...
- **PRODUCT TYPE** = e.g.: 16V4000M93 for EQUI TYPE "Engine", e.g. LM2500 for EQUI TYPE "Gas Turbine"
- **SERIAL NUMBER** = Serial number of a product type
- **BOM-No.** = Bill of material number
- **FUNCTION GROUP** or **FG** = Classification feature of the product according to works standard MTN 5153
- **FG/FGS** = Function Group/Function Group Segment
- **FG DESIGNATION**= Designation of the Function Group according to works standard MTN5153
- **GROUP NAME** = Designation of the bill of material

Table columns

The table contains the following columns depending on the product type.

-  Refers to the catalog page showing a detailed illustration of the part.
- **POS.** (position number): Provides a cross-reference between illustration and text section.
- **PART NO.:** This is the valid number for ordering the part in question (SAP material number).
- **DESIGNATION:** The precise mtu part name (description of the material in SAP).
- **MODEL/TYPE:** Is an additional specification.
- **QTY.** (quantity in unit):
Indicates the quantity of this part required for the installation position shown in the illustration.
Quantities not stated are available on request.
- **UNITS:** The unit of the specified quantity.
PC=Piece, KG= Kilogram, M= Meter, ROL= Roll, PAK= Pack
- **WEIGHT:** Weight of the part in question.
- **FN** (footnote): Provides additional parts information. For clarification of footnote symbols, see nest page.
- **MEASUREMENT:**
The dimensions of the part in question.
For purchase parts, this column may contain the name of the manufacturer.
- **STANDARD:** The applicable standard for the part in question.

Key to the footnotes (column "FN")

AB	Only available complete. Repair by manufacturer only.
AC	Not scheduled as a spare part. Availability must be checked.
ACN	Not scheduled as a spare part. Availability must be checked. Component only available complete.
EM	Emission-relevant parts.
EMN	Emission-relevant parts. Component only available complete.
KD	Group balancing required.
KE	Individual part; not scheduled as a spare part.
KF	Calibrated spare part.
KFN	Calibrated spare part. Component only available complete.
M	Spares-related components or parts with different repair sizes. Dimensions are listed in the Tolerances and Wear Limits List.
MN	Spares-related components or parts with different repair sizes. Components only available complete.
MUB	Spares-related components or parts with different repair sizes. Available only as complete set.
NN	Component is not broken down into further sub-components.
P	List of bought-in components.
SB	Group of parts which are welded or brazed together
UA	Indicate dimensions when ordering
UB	Only supplied complete or as a complete set.

Ordering procedure

The following information is essential for spare-part order processing:

- Product type
- Serial No. (see nameplate)
- FG/FGS
- Item
- Part number
- Designation
- Model/Type
- Quantity
- Page

When ordering spare parts for ancillary equipment such as injection pump, fuel delivery pump, etc., give both model and part number of the part.

When ordering a calibrated part (marked by "KF", "KFN" in the "FN" column), type the following additional information in the **Comments** field:

If the calibration No. is known (see part's nameplate):

Add the calibration No., preceded by an oblique stroke, as an extension to the part number ("/xxxxx")

e.g.: 0025362810/E1234. Here, the extension "E1234" would be the part's calibration number on the nameplate.

If the calibration No. is not known:

Give a detailed description of the required part's installation location by quoting the complete Device Description Code (e.g. "= 37.1.1 + A022 - F003") plus the part number extension "/77" (e.g.: "5005362810/77")

In this case, the calibrated spare part will be supplied under a new part number.

The Device Description Code is found on the nameplate at the part's installation location, or in the corresponding wiring diagram.

When ordering a non-calibrated adjustable part (marked by "KF", "KFN" in the "FN" column) which should not be calibrated by the factory:

Add extension "/76" to the part number (e.g.: 5005362810/76)

Phone orders should be confirmed in writing to prevent errors.

Ordering Example

Product Type	Serial No.	FG/FGS	Pos	Part number	Designation	Qty.	Page
16V4000R41	527101836	031 88	5	5270300701	Crankshaft	1	33
		083 05	5	0020922801	Fuel filter spin-on	1	92
Blue Line	V4000M Waterjet	517 00	75	5185301593	Control console	1	53

Factory repairs

Parts shipped to Rolls-Royce Solutions for repair must be marked with the relevant engine serial number (tag or label, for example).

Exchange components (Reman)

Exchange components are parts overhauled by the manufacturer. They have the same service life and quality characteristics as new parts.

Components which are included in the Reman exchange program are marked by the following letters in front of the part number:

For instance: **E 524 010 35 20 = New component in exchange program**
R 524 010 35 20 = Reman component in exchange program

Exchange procedure

With the introduction of exchange components, the so-called core deposit will also be charged for new parts. The core deposit will be indicated separately on the invoice and the respective value will be credited upon return of reusable components.

Further information upon request.

Emission regulations and emission labels

Responsibility for compliance with emission regulations

Modification or removal of any mechanical/electronic components or the installation of additional components including the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Emission control units/systems may only be maintained, exchanged or repaired if the components used for this purpose are approved by the manufacturer. Noncompliance with these regulations will invalidate the design type approval issued by the emissions regulation authorities. The manufacturer does not accept any liability for violations of the emission regulations.

The maintenance schedules of the manufacturer must be observed over the entire life cycle of the product.

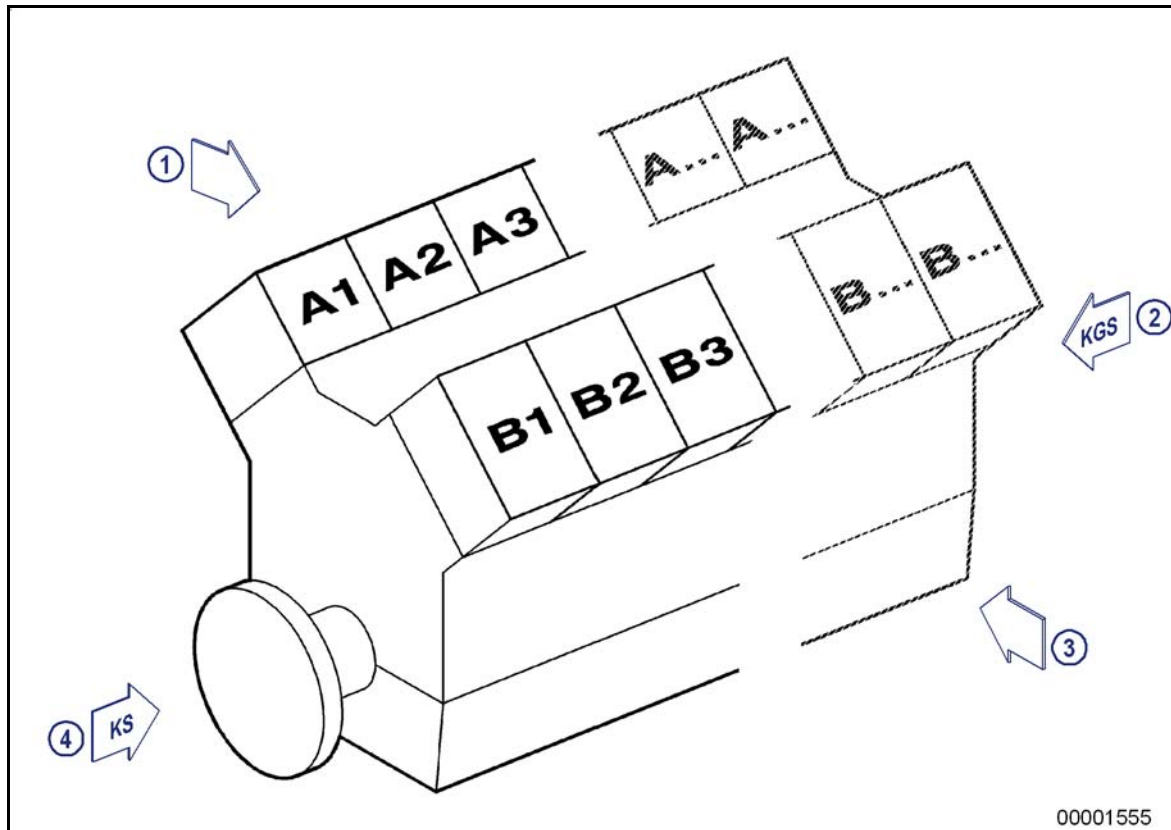
Replacing components with emission labels

Emission labels are attached to all mtu engines. These must remain on the engine throughout its operational life. Engines used exclusively in land-based, military applications other than by US government agencies are excepted from this proviso.

Please note the following when replacing components with emission labels:

- Appropriate emission labels must be affixed on spare parts.
- Emission labels may not be transferred from old components to new ones.
- Emission labels on old components must be removed and destroyed.

Engine side and cylinder designations



- | | |
|---|---|
| 1 Left engine side (A-side) | 3 Right engine side (B-side) |
| 2 Engine free end in accordance with
DIN ISO 1204
(KGS = Kupplungsgegenseite) | 4 Engine driving end in accordance with
DIN ISO 1204 (KS = Kupplungsseite) |

Engine sides are always designated (in accordance with DIN ISO 1204) as viewed from driving end (4).

For cylinder designation (in accordance with DIN ISO 1204) the letter "Ax" refers to the cylinders on the left-hand side of the engine (1) and letter "Bx" refers to the cylinders on the right-hand side (3). The cylinders of each bank are numbered consecutively, starting with x=1 at driving end (4).

The numbering of other engine components also starts with 1 at driving end (4).

FG FG DESIGNATION

004 LABELS AND MISCELLANEOUS MATERIALS

011 CRANKCASE

014 CRANKCASE BOTTOM PART

015 ENGINE MOUNTING

016 END HOUSING, DRIVING END AND FREE END

018 CRANKCASE BREATHER

024 GEAR CASE FREE END

031 CRANKSHAFT WITH BEARING

032 CRANKSHAFT, PTO DRIVING END

035 CRANKSHAFT PTO FREE END

036 TRANSPORT LOCK/LOCK FOR CRANKSHAFT

037 PISTON

038 CONROD

041 CYLINDER HEAD

051 CAMSHAFT

055 VALVE DRIVE

073 INJECTION PUMP / HP PUMP

075 INJECTION VALVE / INJECTOR

077 HP FUEL LINE / HP LINE

081 FUEL LINES BEFORE FUEL PUMP

083 FUEL FILTER

101 EXHAUST TURBOCHARGER, GENERAL OR HP TURBOCHARGER

111 CHARGE-AIR COOLING, GENERAL, LEFT SIDE

121 AIR FILTER

123 AIR SYSTEM

124 AIR PIPING TO CYLINDERS

125 AIR PIPING BETWEEN EXHAUST TURBOCHARGER AND INTERCOOLER

128 CONTROL BLOCK FOR AIR AND EXHAUST FLOW CONTROL FLAPS

142 CONSTANT-PRESSURE MANIFOLD

143 CARRIER HOUSING

144 EXHAUST FLAP WITH ACTUATION

146 EXHAUST LINES AFTER ENGINE (UP TO END OF EGAT)

173 AIR STARTER

181 LUBE OIL PUMP WITH DRIVE

183 OIL FILTRATION/COOLING

185 OIL SUPPLY OF EXHAUST TURBOCHARGERS

202 COOLANT CIRCUIT GENERAL, IF APPLICABLE, HT CIRCUIT

204 HEAT EXCHANGER PLATES

206 COOLANT PIPEWORK WITH THERMOSTAT

207 RAW-WATER PUMP WITH CONNECTIONS

208 EXPANSION TANK

213 BATT.-CHARG. GENERATOR

231 ENGINE MOUNTING

251 STEEL-SPRING COUPLING

255 BENDING-RESILIENT COUPLING

293 OIL PUMP

FG FG DESIGNATION

301 NEUTRAL ACCESSORIES

507 GENERAL CABLING, FOR ENGINE / GEARBOX / GENSET

508 ACCESSORIES F. ENGINE GOVERNOR (ELECTR.) / CONTROL SYSTEM